ST FRANCIS CATHOLIC SECONDARY SCHOOL, LAGOS

The Jesuit-Run Catholic College

Mathematics

JSS 1

First Term

2026/2027

JSS 1 FIRST TERM - Mathematics Scheme of Work

Weeks	Topics	Behavioural Objectives: - At the end of the lesson, students should be able to:
1	Number and numeration	Count, read and write numbers in millions, billions in words.
2	Order of operation	Perform the operation using the four rules
3	Number lines for integers; Directed numbers	Add and subtract of positive and negative numbers Add and subtract any given number correctly
4	Fractions	• Know the meaning and some notations of fraction, Interpret and relate positive and negative numbers to everyday activities. • Addition and subtraction of fractions –Word problems involving addition and subtraction of fractions
5	Multiplication and division of fractions.	• Word problems involving multiplication and division of fractions. • Solve problems on multiplication of fractions.
6.	Conversion of fractions to percentage and vice – versa.	• Word problems on percentage • Solve problems on conversion of fractions.
7	Mid-Term Break	
8	Prime numbers and factors, LCM and HCF of whole numbers	• Identify common factors of whole numbers. • binary numbers, number base, addition and subtraction, basic operation.
9	Rounding off numbers –	• Estimate the dimensions and distances within the school. • Rules for rounding off decimal places, significant figures, nearest tens, thousands, tenths, etc.
10-11	Revision/ Examination	

Date	
Topic	Number and numeration
Objectives/learning outcomes	Understand and work with large numbers (millions, billions)
Insight	I learned to read and write large numbers accurately, including millions and billions.
Key words/New terms	Million, billion, numeration
Application	Reading financial reports or understanding population statistics.
Summary	To understand and work with large numbers involving millions, billions, etc.
What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	
Action goal	

Week One

NUMBER AND NUMERATION

Mathematics began with the ancient people through counting and measuring objects, shapes, heaps and so on.

If a farmer has about five thousand heaps on his farm and needs to know exactly how many. It might be difficult for him to count all at once. He can decide to count in fives, tens, twenties, fifties or hundreds.

In Junior Secondary Schools, we should be able to write and count correctly numbers to a large extent.

Recall that:

Numerals	Words
1 000	One thousand
10 000	Ten thousand
100 000	One hundred thousand
1 000 000	One million
10 000 000	Ten million
100 000 000	One hundred million
1 000 000 000	One billion
100 000 000 000	One hundred billion
1 000 000 000 000	One trillion

NOTE: Students should be able to read 1 – 20 in Yoruba, Igbo and Hausa languages.

BASES OF COUNTING

When people group numbers in five, we say that they are using a base five method of counting. Most people use base ten when counting. For this reason, base ten is used internationally.

Base ten: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

BASE SEVEN

There are seven (7) days in a week. Base seven is used when adding numbers of days and weeks.

Example 1

A baby is 3 weeks and 4 days old. What is its age in days?

Solution 1

3 weeks = $3 \times 7 = 21$

3 weeks and 4 days = 21 days + 4 days

25 days.

Example 2

Find the total of 1 week 6 days, 6 days and 3 weeks 4 days. Give the answer in:

(a)Weeks and days

(b) Days only

Solution 2

(a)In weeks = 1 + 0 + 3 = 4 weeks

In days = 6 days + 6 days + 4 days = 16 days

Converting 16 days to weeks and days = $\frac{16}{7} = 2$ Remainder 2 = 2 weeks and 2 days

Total of weeks and days = 4 weeks + 2 weeks 2 days

= 6 weeks 2 days.

(b) In days only = 6 weeks 2 days

= (6 × 7) days + 2 days

= 42 days + 2 days

= 44 days

Example 3

Suppose today is Tuesday. What day of the week will it be in 150 days' time?

Solution 3

We know that $\frac{150}{7} = 21$ Remainder 3

= 21 weeks 3 days

Next 21 weeks will be on Tuesday.

Then count 3 days further from Wednesday i.e. Wednesday, Thursday and Friday.

Therefore, it will be on Friday.

BASE SIXTY

Base sixty is used when counting in hours, minutes and seconds. 60 seconds make 1 minute, 60 minutes make 1 hour.

Example 4

Find the number of seconds in 3 minutes 49 seconds.

Solution 4

Number of seconds in 3 minutes = 3×60 seconds = 180 seconds

Number of seconds in 3 minutes 49 seconds = 180 seconds + 49 seconds

= 229 seconds

Example 5

Add the following time together. Give the answer in hours and minutes: 3h 40min, 2h 25min, 28min, and 1h 35min.

Solution 5

Add the time in hours together = 3h + 2h + 0h + 1h = 6h

Add the time in minutes together = 40min + 25min + 28min + 35min = 128min

Converting 128min to hours and minutes = $\frac{128}{60} =$ 2 Remainder 8

2h 8min

Total time = 6h + 2h 8min

= 8h 8min.

ROMAN SYSTEMS (NUMERALS)

Despite the fact that the Roman system is the ancient method of writing numbers, the Roman system is still relevant and still in use today.

The table below shows commonly used Roman figures:

Numbers	Roman Numerals
1	I
2	II
3	III
4	IV
5	V
6	VI
7	VII
8	VIII
9	IX

10	X
20	XX
30	XXX
40	XL
50	L
60	LX
70	LXX
80	LXXX
90	XC
100	C
110	CX
150	CL
400	CD
500	D
600	DC
900	CM
1 000	M
2 000	MM
3 000	MMM

Example 6

Write 2 732 in Roman numerals.

Solution 6

2 000 = MM
700 = DCC
32 = XXXII
2 732 = MMDCCXXXII

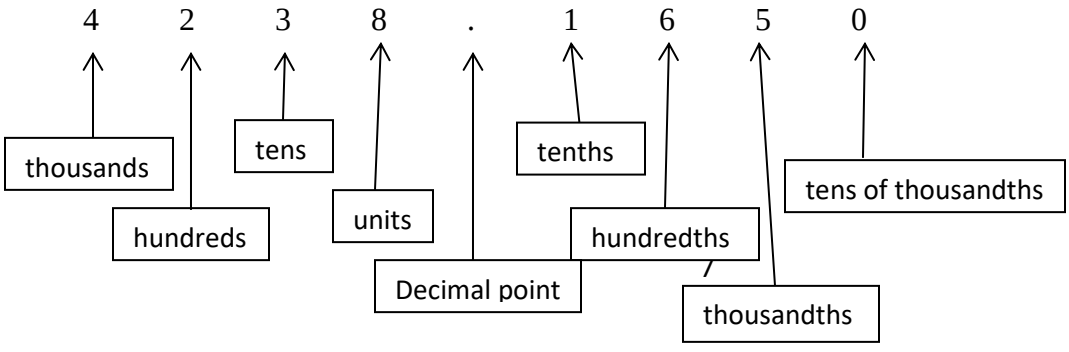
Example 7

What does MCM represent?

Solution 7

M = 1 000
CM = 900
MCM = 1 900

PLACE VALUE SYSTEM



Example 8

Given the number 34 059.712, what is the value of the following:

(a) the 9 (b) the 5 (c) the 0 (d) the 4

(e) the 2 (f) the 3 (g) the 1 (h) the 7

Solution 8

(a) 9 units
(b) 5 tens
(c) 0 hundred
(d) 4 thousands
(e) 2 thousandths
(f) 3 tens of thousands

(g) 1 hundredth

Date	
Topic	Order of Operations
Objectives/learning outcomes	Understand and apply the order of operations (BODMAS)
Insight	I learned to perform mathematical operations in the correct order: Parentheses/Brackets, Exponents/Orders, Multiplication and Division, and Addition and Subtraction.
Key words/New terms	BODMAS, order of operations
Application	I'll apply this skill to solve mathematical problems accurately, including real-life situations like calculating costs or measurements.
Summary	To understand and apply the order of operations using the BODMAS
What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	
Action goal	

Week Two

ORDER OF OPERATIONS

The four basic order of operations of numbers and letters are addition, subtraction, multiplication and division. That is, +, −, ×, ÷

Multiplication

6p is the short form of 6 × p

pq is the short form of p × q

Also, x^4 is the short form of x × x × x × x

Example 1

Simplify the following:

(a) 6ab × $4b^2$

(b) $x^2yz \times yz^2$

Solution 1

(a) 6ab × $4b^2$ = $6 \times 4 \times a \times b \times b \times b$

= 24ab^2

(b) $x^2yz \times yz^2 = x \times x \times y \times y \times z \times z \times z$

$= x^2y^2z^3$

Division

In algebra, letters stand for numbers. Fractions can be reduced to their lowest term by dividing the numerator by the denominator.

Example 2

Simplify the following:

(a) $\frac{2}{3}\ of\ 81y$

(b) $\frac{1}{4}\ of\ x$

(c) $24x^2y \div 3xy$

Solution 2

(b) $\frac{1}{4}\ of\ x = \frac{1}{4} \times x = \frac{x}{4}$

(c) $24x^2y \div 3xy = \frac{24x^2y}{3xy} = \frac{24\times x\times x\times y}{3\times x\times y} = 8x$

ORDER OF OPERATION

B – Bracket

O – Of

D – Division

M – Multiplication

A – Addition

S – Subtraction

Example 3

Simplify the following:

(a) $18 - 6 \times 2 + 15 \div 3$

(b) $16a \div 2 - 3a \times 2 - 15a \div 3$

(c) $6p \times 2 + 2p - 40p \div 4$

Solution 3

NOTE: While using **BODMAS**, **M**ultiplication and **D**ivision can be used simultaneously at the same time.

a.)$18 - (6 \times 2) + (15 \div 3) = 18 - 12 + 5$

$= 18 + 5 - 12 = 23 - 12 = 11$

b.) $(16a \div 2) - (3a \times 2) - (15a \div 3)$

$= 8a - 6a - 5a = 8a - 11a$

$= -3a$

c.) $(6p \times 2) + 2p - (40p \div 4) = 12p + 2p - 10p$

$= 14p - 10p = 4p$

PERIOD II

REMOVING BRACKETS

Positive Sign Before A Bracket

The positive sign before a bracket will open-up the bracket through multiplication.

Example 5

Simplify the following:

a.) $8x + (3x + 4y)$

b.) $(3p - 4q) + (2p - 8q)$

Solution 5

a.) $8x + 3x + 4y = 11x + 4y$

b.) $3p - 4q + 2p - 8q = 3p + 2p - 4q - 8q$

$= 5p - 12q$

Negative Sign Before A Bracket

The negative sign before a bracket will open-up the bracket through multiplication.

Example 6

Simplify the following:

a.) $10y - (6y - 4x)$

b.) $(5a + 10b) - (2a + 15b)$

c.) $(2x - 3y) - (-5x + 6y)$

Solution 6

a.) $10y - (6y - 4x) = 10y - 6y + 4x$

$= 4y + 4x$ or $4(y + x)$

b.) $(5a + 10b) - (2a + 15b) = 5a + 10b - 2a - 15b$

$= 5a - 2a + 10b - 15b$

$= 3a - 5b.$

c.) $(2x - 3y) - (-5x + 6y) = 2x - 3y + 5x - 6y$

$= 2x + 5x - 3y - 6y$

$= 7x - 9y$

Example 7

Simplify the following:

a.) $(7 - 10) - (-12 + 8)$

b.) $-6a + (3a - 7b) - (10a - 15b)$

Solution 7

a.)Using opening of bracket

$7 - 10 + 12 - 8 = 19 - 18 = 1$

b.) $-6a + 3a - 7b - 10a + 15b = -6a + 3a - 10a - 7b + 15b$

$= -13a + 8b$ or $8b - 13a$.

Word Problems Involving Order Of Operation

Example 8

A Pen cost h naira and a pencil cost ₦ 30.00 less than a pen. Find the cost of a pen and a pencil.

Solution 8

Cost of a pen = ₦ h

Cost of a pen = ₦ $(h - 30)$

$= ₦ h + h - 30$

$= ₦ (2h - 30)$

Additional Example 9

The greater of two consecutive numbers is x.

a.) Find the sum of two numbers.

b.) Subtract the sum of the two numbers from 5x

Solution

a.)If x is the greater of two consecutive numbers, the lower number is $x-1$

Sum of the two numbers = $x+(x-1)$

$= x + x - 1$

$= 2x - 1$

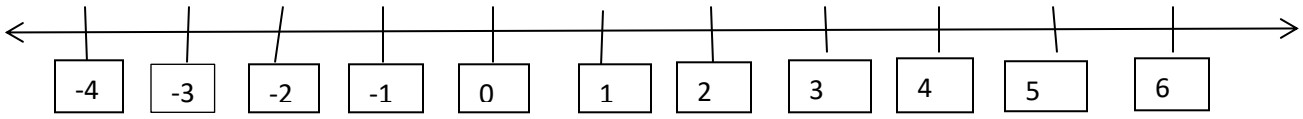
b.) $5x - (2x-1) = 5x - 2x + 1 = 3x + 1$

Date	
Topic	Number Line for Integers
Objectives/learning outcomes	Understand and visualize integers on a number line
Insight	I learned to represent integers (positive, negative, and zero) on a number line, understanding their relationships and ordering.
Key words/New terms	Integers, number line, positive, negative, zero
Application	I'll apply this skill to solve problems involving temperature, elevation, or financial transactions, where integers are used.
Summary	To understand and visualize integers on a number line
What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	
Action goal	

WEEK THREE

NUMBER LINE FOR INTEGERS

A number line is a picture of the counting numbers. It starts from zero (0) and increases or decreases in equal steps to the right or the left .The number line can be used to show addition and subtraction.



The positive numbers are marked to the right of 0 (zero) and negative numbers are marked to the left of zero (0).

Integers are the positive and negative whole numbers including zero (0).

$Z = [\ldots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \ldots]$

Example 1

Arrange the following numbers in descending order:

$-5, 10, -8, 0, 5, 2, 7, -3$

Solution 1

Descending order means from highest to lowest

$= 10, 7, 5, 2, 0, -3, -5, -8.$

Addition and Subtraction Using Number Line

Example 2

Simplify the following using number line:

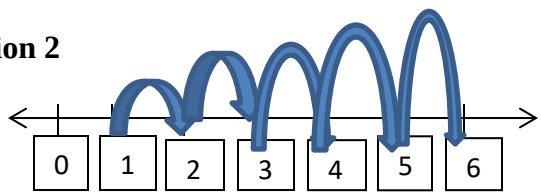
a.) $1+5$

b.) $6 - 3$

c.) $5 - 7$

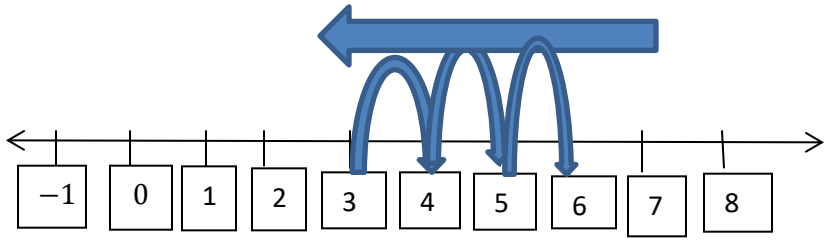
Solution 2

a.)



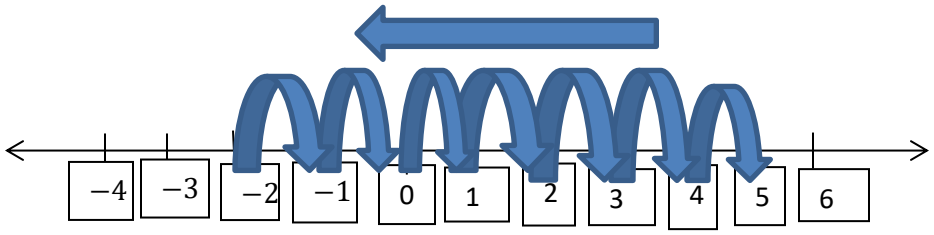
$\therefore 1 + 5 = 6$

b.)



Therefore $6 - 3 = 3$

c.)



Therefore, $5 - 7 = -2$

Example 3

Simplify the following using number line:

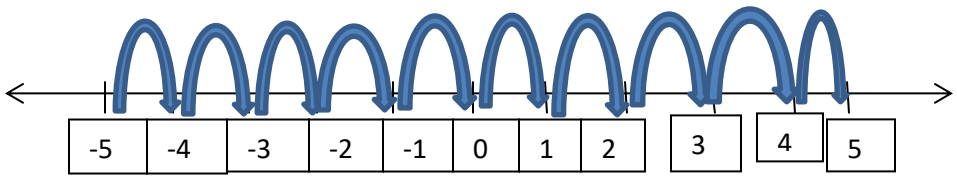
a.) $-5 + 10$

b.) $-3 - 4$

c.) $2 - 7$

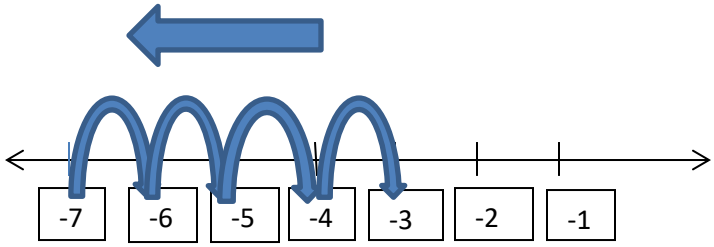
Solution 3

a.)



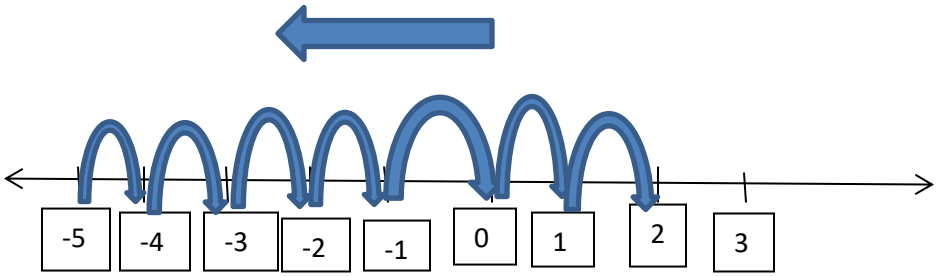
Therefore $-5+5=5$

b.)



Therefore -3+4 = -7

c.)



Therefore 2-7 = -5.

Date	
Topic	Addition and Subtraction of Positive and Negative Numbers
Objectives/learning outcomes	Understand and apply rules for adding and subtracting positive and negative numbers.
Insight	- Adding two positives: positive result - Adding two negatives: negative result - Adding positive and negative: subtract absolute values, keep sign of larger - Subtracting a negative: same as adding a positive
Key words/New terms	Positive numbers, negative numbers, absolute value
Application	I'll apply this skill to solve real-life problems, such as calculating temperatures, debts, or elevations.
Summary	To understand and apply rules for adding and subtracting positive and negative numbers.
What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	
Action goal	

WEEK FOUR

ADDITION AND SUBTRACTION OF POSITIVE AND NEGATIVE NUMBERS

Example 1

Simplify the following:

a.) $3 + 4 = 7$

b.) $-3 + 4 = 1$

c.) $-3 - 5 = -8$

d.) $12 - 17 = -5$

e.) $3 - 11 + 2 = 3 + 2 - 11 = 5 - 11 = -6$

Example 2

Simplify the following:

a.) $19 - 11$

b.) $11 - 19$

c.) $-11 - 19$

d.) $-33 - 12$

e.) $78 - 252$

f.) $16 + (-9) - (-10)$

g.) $(-12) + (-8) + 15$— 5.

Solution 2

a.)$19 - 11 = 8$

b.) $11 - 19 = -8$

c.) $-11 - 19 = -30$

d.) $-33 - 12 = -45$

e.) $78 - 252 = -174$

f.) $16 + (-9) — 10 = 16 - 9 + 10 = 16 + 10 - 9 = 26 - 9 = 17$

g.) $(-12) + (-8) + 15 — 5 = -12 - 8 + 15 + 5$

$= 15 + 5 - 12 - 8 = 20 - 20 = 0$ Or

$= -20 + 20 = 0.$

Example 3

a.) What must be added to 26 to make -4

b.) What must be subtracted from -3 to make 4.

Solution 3

a.) Let the number to be added be x

$26 + x = -4$

$x = -4 - 26$

$x = -30$

Therefore -30 must be added to 26 to make -4.

b.) Let the number to be subtracted be q

$-3 - q = 4$

$-3 - 4 = q \;\; ; \;\; -7 = q \; or \; q = -7$

$Therefore$ -7 must be subtracted from -3 to make 4.

Date	
Topic	Fractions
Objectives/learning outcomes	Understand and work with fractions, including simplifying, adding, subtracting, multiplying, and dividing.
Insight	I learned that fractions represent part-whole relationships and can be manipulated using various operations.
Key words/New terms	Numerator, denominator, equivalent fractions, simplest form
Application	I'll apply fraction skills to real-life situations, such as cooking, measurement, or finance.
Summary	To understand and work with fractions, including simplifying, adding, subtracting, multiplying, and dividing.
What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	
Action goal	

WEEK FIVE

FRACTIONS

A fraction represents part of a whole quantity. A fraction is in two parts.

i) Numerator: This is the number at the top in a fraction.

ii.) Denominator: This is the number below in a fraction.

For instance: $\frac{3}{7} = \frac{numerator}{denominator}$

Types of fraction

i) **Proper Fraction**: Is the fraction with numerator less than the denominator.

e.g. $\frac{2}{3}, \frac{1}{2}, \frac{4}{5}, etc$

ii) **Improper fraction**: Is the fraction with the numerator greater than the denominator.

e.g. $\frac{7}{4}, \frac{10}{3}, \frac{13}{2}, etc$

iii) **Mixed fraction**: Is the type of fraction which involves the combination of whole number and fraction.

e.g. $3\frac{1}{2}, 2\frac{2}{3}, 7\frac{1}{2}, etc.$

CONVERSION OF FRACTION

Example 1

Convert the following fraction to decimal fractions:

(a)$\frac{2}{5}$ (b) $\frac{1}{3}$

Solution 1

a) $\frac{2}{5}$ = (using the long division) 0.4 (terminating decimal)

b) $\frac{1}{3}$ = (using the long division) 0.333… (Non-terminating decimal or recurring decimal)

Hence, $\frac{1}{3}$ = 0.33 (to 2 d.p.) Or 0.333 (to 3 d.p.)

Example 2

Convert the following from decimal fraction to common fraction:

(a) 0.25 (b) 0.4

Solution 2

a) 0.25 = $\frac{25}{100} = \frac{1}{4}$

b) 0.4 = $\frac{4}{10} = \frac{2}{5}$

Example 3

Arrange the fraction $\frac{3}{4}, \frac{5}{8}, \frac{7}{12}, \frac{2}{3}$ in:

(a) ascending order

(b) descending order

Solution 3

LCM of the denominators = 24

$$= \frac{3\times6,5\times3,7\times2,2\times8}{24}$$

$$= \frac{18,15,14,16}{24}$$

a) Ascending order (lowest to highest) = $\frac{7}{12}, \frac{5}{8}, \frac{2}{3}, \frac{3}{4}$

b) Descending order (highest to lowest) = $\frac{3}{4}, \frac{2}{3}, \frac{5}{8}, \frac{7}{12}$

Example 4

Find the missing number in the box below: $\frac{2}{9} = \frac{\quad}{12}$

Solution 4

Let the missing number = x

Hence, $\frac{2}{9} = \frac{x}{12}$

9x = 2 × 12; 9x = 24 Divide both sides by 9

x = $\frac{24}{9} = \frac{8}{3}$ or $2\frac{2}{3}$

PERIOD II

ADDITION AND SUBTRACTION OF FRACTION

Example 5

Simplify the following fractions:

$a) \frac{5}{6} + \frac{3}{8}$

b) $\frac{7}{10} - \frac{4}{15}$

c) $1\frac{1}{2} + 2\frac{2}{3}$

Solution 5

a) $5\frac{5}{6} + \frac{3}{8} = \frac{20+9}{24} = \frac{29}{24} = 1\frac{5}{24}$

b) $\frac{7}{10} + \frac{4}{15} = \frac{21-8}{30} = \frac{13}{30}$

c) $1\frac{1}{2} + 2\frac{2}{3} = 3\frac{3+4}{6} = 3\frac{7}{6} = 3 + 1\frac{1}{6} = 4\frac{1}{6}$

Example 6

Simplify the following fractions:

a) $6\frac{5}{12} - 3\frac{3}{4} + 2\frac{2}{3}$ b) $7\frac{1}{5} + 1\frac{1}{4} - 5\frac{4}{5}$

Solution 6

a) $6\frac{5}{12} - 3\frac{3}{4} + 2\frac{2}{3} = 5\frac{5-9+8}{12} = 5\frac{5+8-9}{12} = 5\frac{4}{12} = 5\frac{1}{3}$

b) $7\frac{1}{5} + 1\frac{1}{4} - 5\frac{4}{5} = 3\frac{4+5-16}{20} = 3\frac{9-16}{20} = 2\frac{20+9-16}{20} = 2\frac{13}{20}$

WORD PROBLEMS INVOLVING ADDITION AND SUBTRACTION OF FRACTIONS

Example 7

A man goes on a journey. He does $\frac{1}{6}$ of it on a bicycle, $\frac{4}{5}$ on a lorry and walks the rest. What fraction of the journey does he walk?

Solution 7

The total journey as a whole = 1

Journey on bicycle and lorry = $\frac{1}{6} + \frac{4}{5} = \frac{5+24}{30} = \frac{29}{30}$

Therefore, the fraction of the journey walked by the man = $\frac{1}{1} - \frac{29}{30} = \frac{30-29}{30} = \frac{1}{30}$

OR $\frac{1}{1} - \left(\frac{1}{6} + \frac{4}{5}\right) = \frac{1}{1} - \left(\frac{5+24}{30}\right) = \frac{1}{1} - \frac{29}{30} = \frac{1}{30}$

Date	
Topic	Fractions
Objectives/learning outcomes	Understand and apply the concept of multiplying fractions.
Insight	I learned that multiplying fractions involves multiplying the numerators and multiplying the denominators.
Key words/New terms	Numerator, denominator, equivalent fractions, simplest form
Application	I'll apply fraction multiplication to real-life situations, such as scaling recipes or calculating areas
Summary	To understand and apply the concept of multiplying fractions.
What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	
Action goal	

WEEK SIX

MULTIPLICATION AND DIVISION OF FRACTION

Example 1

Simplify the following: (a) $9 \times \frac{4}{3}$ (b) $\frac{2}{3}\ of\ 30$

(c) $4\frac{5}{8} \times \frac{4}{25}$ (d) $\frac{5\times21}{14\times15}$

Solution 1

(a)$\frac{9}{1} \times \frac{4}{3} = 12$

(b) $\frac{2}{3} of\ 30 = \frac{2}{3} \times \frac{30}{1} = 20$

(c) $4\frac{5}{8} \times \frac{4}{25} = \frac{37}{8} \times \frac{4}{25} = \frac{37}{50}$

(d) $\frac{5\times21}{14\times15} = \frac{1}{2}$

Example 2

Simplify the following: (a) $1\frac{2}{3} \div 5$ (b) $6 \div \frac{3}{5}$ (c) $\frac{12}{25} \div \frac{9}{10}$

(d) $\frac{3}{10} \times \frac{35}{36} \div \frac{14}{15}$ (e) $\frac{7\frac{3}{7}}{9\frac{3}{4}} \div \frac{5}{21}$

Solution 2

a) $1\frac{2}{3} \div 5 = \frac{5}{3} \div \frac{5}{1} = \frac{5}{3} \times \frac{1}{5} = \frac{1}{3}$

b) $6 \div \frac{3}{5} = \frac{6}{1} \times \frac{5}{3} = 10$

c) $\frac{12}{25} \div \frac{9}{10} = \frac{12}{25} \times \frac{10}{9} = \frac{8}{15}$

d) $\frac{3}{10} \times \frac{35}{36} \div \frac{14}{15} = \frac{3}{10} \times \frac{35}{36} \times \frac{15}{14} = \frac{5}{16}$

e) $\frac{7\frac{3}{7}}{9\frac{3}{4}} \div \frac{6}{21} = \frac{52}{7} \div \frac{39}{4} \div \frac{5}{21} = \frac{52}{7} \times \frac{4}{39} \times \frac{21}{5} = \frac{16}{5} = 3\frac{1}{5}$

PERIOD II

WORD PROBLEMS INVOLVING MULTIPLICATION AND DIVISION OF FRACTIONS

Example 3

One lecture at an evening course lasts $1\frac{1}{2}$ hours. If the course last 30 hours altogether, how many lectures are there?

Solution 3

One lecture last $1\frac{1}{2}$ hours

Total time for the course = 30 hours

1 lecture = $1\frac{1}{2}$ hours

x lectures = 30 hours

1 lecture = $\frac{3}{2}$ hours

x lectures = 30 hours

Therefore, $\frac{3x}{2} = 30$

$3x = 30 \times 2$

$x = \frac{30\times2}{3} = 20\ lectures$

Example 4

Three sisters share some money. The oldest get $\frac{5}{11}$ of the money. The next girl gets $\frac{7}{12}$ of the remainder. What fraction of the money does the youngest girl gets?

Solution 4

The oldest sister gets $\frac{5}{11}$ of the money. The next girl gets $\frac{7}{12}$ of the remainder.

Remainder = $\frac{1}{1} - \frac{5}{11} = \frac{11-5}{11} = \frac{6}{11}$

Therefore, $\frac{7}{12}$ of remainder = $\frac{7}{12} \times \frac{6}{11} = \frac{7}{22}$

The next girl gets $\frac{7}{22}$ of the money.

The fraction of the money got by the youngest girl $= \frac{1}{1} - \left(\frac{5}{11} + \frac{7}{22}\right)$

$= \frac{1}{1} - \left(\frac{10+7}{22}\right)$

$= \frac{1}{1} - \frac{17}{22} = \frac{22-17}{22} = \frac{5}{22}$

Date	
Topic	Percentages and Fractions
Objectives/learning outcomes	Understand and convert between percentages and fractions.
Insight	I learned that percentages can be represented as fractions with a denominator of 100.
Key words/New terms	Percentage, fraction, decimal, conversion
Application	I'll apply percentage and fraction skills to real-life situations, such as calculating discounts, interest rates, or statistics.
Summary	To understand and convert between percentages and fractions.
What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	
Action goal	

Week Eight

PERCENTAGES AND FRACTIONS

Percentage is a special type of fraction which can be expressed as either a common fraction or a decimal fraction.

The symbol %is used in place of the word percentage.

29% means 29 hundredths, i.e. 29% = $\frac{29}{100}$ or 0.29

We can therefore conclude that percent is a fraction with 100 as the denominator.

Example 1

Express 15% as a fraction in its lowest term.

Solution 1

15% = $\frac{15}{100} = \frac{3}{20}$

Example 2

Express the following fraction as percentage: (a) $\frac{2}{5}$ (b) $\frac{176}{300}$

Solution 2

(a)$\frac{2}{5} \times 100\% = 40\%$

(b) $\frac{175}{300} \times 100\% = \frac{176}{3}\% = 58\frac{2}{3}\%\ or\ 58.67\%$

Example 3

Change the following percent to decimals: (a) 75% (b) 7.2%

Solution 3

(a) 75% = $\frac{75}{100} = 0.75$

(b) 7.2% = $\frac{7.2}{100} = 0.072$

Example 4

Express 7min 30sec. as a fraction of 1hour.

Solution 4

NOTE: Convert both quantities of time to the same unit

i.e. 7min 30sec. = $(7\times 60 + 30)\ seconds$

= 450 seconds

1 hour = $60 \times 60 = 3\,600\ seconds$

Therefore, $\frac{450}{3600} = \frac{1}{8}$

PERIOD II

WORD PROBLEMS ON PERCENTAGE

Example 5

A motorist buys a 5 litre can of gear oil. She used 800ml. What percentage of oil has she used and what percentage remains?

Solution 5

Convert to same unit either ml or L

1 000ml = 1 litre

800ml = x litres

1 000x = 800 Divide both sides by 1 000

x = $\frac{800}{1\,000} = 0.8\ litre$

Therefore, the motorist used 0.8 litre of oil

Percentage of oil used = $\frac{litres\ of\ oil\ used}{total\ litres\ of\ oil} \times 100\% = \frac{0.8}{5} \times 100\% = 16\%$

Percentage of oil remaining i.e. not used = (100 – 16) %

= 84%

Example 6

A worker hammers a post 2.2m long into the ground 66cm of the post is below the ground. What percentage of the post is above the ground?

Solution 6

Convert both measurements to same unit.

Length of post 2.2m long = $(2.2 \times 100)cm$

= 220cm

Length of the post above the ground = (220 – 66) cm

= 154cm

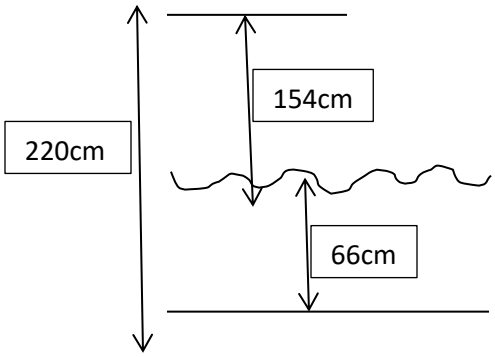
Therefore, percentage of the post above the ground

= $\frac{length\ of\ post\ above\ the\ ground}{total\ length\ of\ the\ post} \times 100\%$

= $\frac{154}{220} \times 100\% = 70\%$

Therefore, percentage above the ground = 70%

Percentage below the ground = (100 – 70)% = 30%



Date	
Topic	Factors and Multiples
Objectives/learning outcomes	Understand and identify factors and multiples of numbers.
Insight	I learned that factors are numbers that divide a given number exactly, while multiples are products of a given number and an integer.
Key words/New terms	Factor, multiple, prime factorization
Application	I'll apply factor and multiple skills to real-life situations, such as simplifying fractions, finding common denominators, or solving problems involving patterns.
Summary	To understand and identify factors and multiples of numbers.
What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	
Action goal	

Week Nine

FACTORS AND MULTIPLES

FACTORS

A factor is a number that can divide a given number without remainder.

Factors of 40 are: 1, 2, 4, 5, 8, 10, 20 and 40

Example 1

Find all the factors of63.

Solution 1

Factors of 63 are: 1, 3, 7, 9, 21 and 63

PRIME NUMBERS

A prime number has only two factors; itself and 1.

Prime numbers are: 2, 3, 5, 7. 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47 …

1 is not a prime number because it has only one factor, itself. 9 is not a prime number because it has more than two factors: 1, 3 and 9.

The prime factors of a number are the factors of the numbers that are prime.

Example 1

Express 15, 24 and 42 as the product of their prime factors.

Solution 1

$15 = 3\times 5$

$24 = 2\times 2 \times 2 \times 3$

$42 = 2\times 3 \times 7$

Example 2

Express as a product of primes in index form: (a) 24 (b) 18

Solution 2

(a) $24 = 2\times 2 \times 2 \times 3 = 2^3 \times 3$

(b) $18 = 2\times 3 \times 3 = 2 \times 3^2$

HIGHEST COMMON FACTOR (HCF)

Example 3

Find the HCF of 18, 24 and 42

Solution 3

18 = $\boldsymbol{2} \times \mathbf{3} \times 3$

24 = $\boldsymbol{2} \times 2 \times 2 \times \boldsymbol{3}$

42 = $\boldsymbol{2} \times \mathbf{3} \times 7$

Therefore, HCF = $\mathbf{2 \times 3} = 6$

Method 2

2	18	24	42
3	9	12	21
	3	4	7

Since there is no more common factors;

Therefore, HCF = $2 \times 3 = 6$

MULTIPLES

6 is a factor of 12, 18, 24, 30, 36 …

We say that 12, 18, 24, 30, 36 … are all multiples of 6.

Example 4

Give five multiples of 20

Solution 4

Multiple of 20: 40, 60, 80, 100, 120.

LOWEST COMMON MULTIPLES (LCM)

30 is the lowest common multiple (LCM) of 6, 10 and 15.

Example 5

Find the lowest common multiple (LCM) of 24, 28 and 50

Solution 5

$24 = 2 \times 2 \times 2 \times 3 = 2^3 \times 3$ (index form)

$28 = 2 \times 2 \times 7 = 2^2 \times 7$ (Index form)

$50 = 2 \times 5 \times 5 = 2 \times 5^2$ (Index form)

LCM = $2 \times 2 \times 2 \times 3 \times 5 \times 5 \times 7$

$=2^3 \times 3 \times 5^2 \times 7 = 4200$

NOTE: The prime factors are 2, 3, 5 and 7. The LCM contains the highest power of 2, 3, 5 and 7 that appear in the factorized numbers.

Method 2

LCM of 24, 28 and 50

2	24	28	50
2	12	14	25
2	6	7	25
3	3	7	25
5	1	7	25
5	1	7	5
7	1	7	1
	1	1	1

Therefore LCM = $2 \times 2 \times 2 \times 3 \times 5 \times 5 \times 7 = 4200$

PERIOD II

ROUNDING OFF NUMBERS

Rules For Rounding Off

When rounding off the numbers, we round up or round down depending on the digit being rounded. This digit is always the nearest to the digit to be rounded off.

We round up numbers between 5-9 and the number rounded up becomes 1 which is added to the rounded digit.

We round down numbers between 0 - 4 and the number rounded down becomes 0, which is added to the rounded digit.

Decimal Places and Significant Figures

Example 6

Round off 17.8275 to

i.)2 decimal places (d.p)

ii) 2 significant figures (s.f)

iii) 1 d.p

iv) 3 s.f

v) Nearest whole number.

vi) Nearest tens

vii) Nearest thousandths

Solution 6

i) $17.8275 \approx 17.82$ (to 2 d.p)

ii) $17.8275 \approx 18$ (to 2 s.f)

iii) $17.8275 \approx 17.8$ (to 1 d.p)

iv) $17.8275 \approx 17.8$ (to 3 s.f)

v) $17.8275 \approx 18$ (to nearest whole number)

vi) $17.8275 \approx 20$ (to nearest tens)

vii) $17.8275 \approx 17.8275$ (to nearest thousandths/ 3 d.p)

Example 7

Round off 0.003459 to

i)2 d.p

ii.) 2 s.f

iii.) nearest whole number

iv.) nearest hundredth

v.) 4 d.p

Solution 7

i.)0.003459 ≈ 0.00 (to 2 d.p)

ii.) 0.003459 ≈ 0.0035 (to 2 s.f)

iii.) 0.003459 ≈ 0 (to the nearest whole number)

iv.) 0.003459 ≈ 0.00 (to nearest hundredths)

v.) 0.003459 ≈ 0.0035 (to 4 d.p)

Example 8

Approximate 79.75 to the nearest.

a.)Hundred

b.) Tens

c.) Whole number

d.) Tenths

Solution 8

a.)79.75 ≈ 100 (to nearest hundredths)

b.) 79.75 ≈ 80 (to nearest tens)

c.) 79.75 ≈ 80 (to nearest whole number)

d.) 79.75 ≈ 79.8 (to nearest tenths)

Example 9

Given 14505 to the nearest

a.) Thousands

b.) Hundred.

c.) Tens

Solution 9

a.)14505 ≈ 15000 (to nearest thousands)

b.) 14505 ≈ 14500 (to nearest hundreds)

c.) 14505 ≈ 14510 (to nearest tens)